EXECUTIVE TECHNICAL REPORT (A4 LANDSCAPE)

Triton — Rainwater Harvesting | Project: **Example - tutorial house in New York (ERA5 1991-2020) [costs to be entered by the user]**

Project ref.: **N5-20261015-085015-27731E68** | Issued: **2026-10-15** | **International Edition (INT)**

Normative compliance: **Codes and standards applicable at the project site, to be stated by the responsible engineer**

CLIMATE SITE	CLIMATE DATASET	SERIES PERIOD	ECONOMIC HORIZON
New York / USA	INM-CM4 · HIST	01/01/1991 a 31/12/2020	20 years

RECOMMENDED VOLUME	POTABLE-WATER SAVINGS	NPV (20 YEARS)	ESTIMATED PAYBACK
14,000 L	**28.0%**	**$ 1,343.38**	**0.0 years**
Lower Cistern	Mains-Water Replacement	Net Present Return	0.0 years (discounted)

1. BUILDING & HYDROLOGY

Technical Parameter	Value
Rainfall Series	**NewYork.csv**
Catchment Area	**150.0 m²**
Runoff Coeff. (C)	0.85
First-Flush Discard	2.0 mm
Population Served	4 people
Target Rainwater Demand (28%)	**168 L/day**

2. SIMULATED WATER BALANCE

Operational Indicator	Performance
Cistern Volume	**14,000 L**
Potable-Water Savings	**28.0%**
Mean Rainwater Use	**167.9 L/day**
Mean Potable-Water Use	432.1 L/day
Overflow Volume	137.8 L/day
Full Supply (100%)	**99.9% of days**
No Supply (0%)	**0.1% of days**

3. ECONOMIC FEASIBILITY

Financial Indicator	Result
CAPEX Investment	**$ 0.00**
Gross Savings, Year 1	**$ 300.05 / yr**
Simple Payback	0.0 years
Discounted Payback	0.0 years
NPV (20 Years)	**$ 1,343.38**
IRR (% p.a.)	**0.0% p.a.**

📌 Technical Opinion on Water Engineering & Sustainability

Under the recorded assumptions, the rainwater system shows modelled potable-water savings of **28.0%**; NPV of **$ 1,343.38** and a simple payback of **0.0 years**; under the assumptions adopted, the investment pays for itself within the analysis horizon. Sizing follows the technical standards and regulations applicable at the project site.

⚠️ Technical Responsibility & Validation Notice

Triton is a computational sizing-support tool. **The licensed professional is responsible for** validating the executive assumptions, network interferences and compliance with local legislation.